

MTH:2310 DISCRETE MATH

QUIZ 3

DUE: MONDAY FEB. 24

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: (4 points)

Your ability to express your solutions is just as important as mathematical accuracy.

You will be penalized up to 2 points for arithmetic errors.

You will be penalized up to 2 points for disorganized or hard to read solutions.

Problem 1. (24 points) Rules of Inference

Determine whether the following arguments are valid **and state each argument's corresponding argument form**. Justify why an argument is (or fails to be) valid.

- (a) (8 points) "If it does not rain or if it does not snow, then the disc golf tournament will be held. If the disc golf tournament is held, the trophy will be awarded. The trophy was not awarded. Therefore, it rained."

p = "It rains."
 q = "It snows."
 r = "The disc golf tournament is held."
 s = "The trophy is awarded."

$$(\neg p \vee \neg q) \Rightarrow r$$

$$r \Rightarrow s$$

$$\neg s$$

$$\frac{\quad}{\therefore p}$$

Arg. Form is

$$((\neg p \vee \neg q) \Rightarrow r) \wedge (r \Rightarrow s) \wedge (\neg s) \Rightarrow p$$

This is only false if (premises) = T and $p = F$.

Then $(\neg p \vee \neg q) \Rightarrow r$
 $\equiv (T \vee \neg q) \Rightarrow r$
 $\equiv T \Rightarrow r$
 is only true
 if $r = T$.

$$r \Rightarrow s$$

$$\equiv T \Rightarrow s$$

is true only
 if $s = T$.

$$\neg s$$

$$\equiv F$$

but $\neg s$
 must
 be T
 So this argument
 is valid.

- (b) (8 points) "If Johnathan P. Math works all night on the suggested problems, then he can answer all the quiz questions. If Johnathan P. Math answers all the quiz questions, he will understand the material. Therefore, if Johnathan P. Math work all night on the suggested problems, he will understand the material."

p = "Johnathan P. Math works all night on suggested problems."
 q = "Johnathan P. Math can answer all quiz questions."
 r = "Johnathan P. Math understands the material."

$$\begin{array}{l} p \Rightarrow q \\ q \Rightarrow r \\ \hline \therefore p \Rightarrow r \end{array}$$

Arg. Form is

$$((p \Rightarrow q) \wedge (q \Rightarrow r)) \Rightarrow (p \Rightarrow r)$$

This is valid since it is a
Hypothetical Syllogism

- (c) (8 points) "If x is a positive real number, then x^2 is a positive real number. Therefore, if a^2 is positive where a is a real number, then a is a positive real number."

$P(x)$: " x is positive."
 $Q(x)$: " x^2 is positive."
 where x is a real number.

$$\begin{array}{l} \forall x (P(x) \Rightarrow Q(x)) \\ \hline \therefore Q(a) \Rightarrow P(a) \end{array} \quad \begin{array}{l} \text{Arg. Form is} \\ (\forall x (P(x) \Rightarrow Q(x))) \Rightarrow (Q(a) \Rightarrow P(a)) \end{array}$$

Consider the real number $a = -1$.

$$a^2 = (-1)^2 = 1 \text{ is positive.}$$

but $a = -1$ is negative.

Therefore this argument is invalid by counter example.

Problem 2. (8 points) Rules of Inference

Let the following set of statements be the premises of an argument. Identify four distinct valid conclusions that can be made, and state which rule(s) of inference used which make the conclusion valid.

- $\overset{\text{"x is bug, x is insect", "x has 6 legs"}}{\forall x (P(x) \Rightarrow Q(x))}$
 "All insects have six legs." $\rightarrow \forall x (P(x) \Rightarrow Q(x))$
 "Dragonflies are insects." $\rightarrow P(D)$
 "Spiders do not have six legs." $\rightarrow \sim Q(S)$
 "Spiders eat dragonflies." $\rightarrow R(S, D) \rightarrow S \text{ eats } D$
 "I squish insects unless they can sting." $\rightarrow \forall x (T(x) \Rightarrow U(x))$
- "Dragonflies have six legs." Universal Modus Ponens
 "Spiders are not insects." Universal Modus Tollens
 "There is an insect spiders eat." Existential Generalization
 "I squish dragonflies unless they sting." Universal Instantiation

Problem 3. (8 points) Introduction to Proofs

Use a direct proof to show that if x is any non-zero rational number, then $\frac{1}{x}$ is a rational number.

- $\forall x (P(x) \Rightarrow Q(x))$
 x is a number
 $P(x)$: " x is a non-zero rational number."
 $Q(x)$: " x^2 is a rational number."
- Let n be an arbitrary non-zero rational number.
 Then there are integers p, q ($q \neq 0$) such that
 $n = \frac{p}{q}$. Since $n \neq 0$, $p \neq 0$.
 Notice $\frac{1}{n} = \frac{1}{p/q} = \frac{q}{p}$
 so $\frac{1}{n}$ is a rational number.
 Therefore, if x is a non-zero rational number,
 then $\frac{1}{x}$ is rational. $\square$

Problem 4. (10 points) *Introduction to Proofs*

Use a proof by contraposition to show that if x, y, z are integers and the sum $x + y + z$ produces an odd integer, then at least one of x, y , and z is an odd integer.

$$\forall x, y, z (P(x, y, z) \rightarrow Q(x, y, z))$$

x, y, z are integers.

$P(x, y, z) = "x + y + z \text{ is odd}."$

$Q(x, y, z) = "At least one of x, y, z are odd."$

$\sim Q(x, y, z) = "All of x, y , and z are even."$

$\sim P(x, y, z) = "x + y + z \text{ is even}"$

Let a, b, c be arbitrary integers.

Assume a, b , and c are even. Then there are integers u, v, w such that

$$a = 2u, \quad b = 2v, \quad c = 2w.$$

$$\text{Notice } a + b + c = 2u + 2v + 2w = 2(u + v + w)$$

so $a + b + c$ is even.

Therefore by proof by contraposition, if x, y, z are integers and $x + y + z$ is odd, then at least one of x, y, z are odd. 